

Everyone Was Right: Lexical Representations in Language Models Are Both Abstract and Holistic

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Abstract

Lexical representations are central to linguistic theory, yet to what extent they’re memorized, item-specific representations or abstract, generalizable representations remains a core debate. In the present study, we investigate how Large Language Models (LLMs) navigate this distinction using a quantitative framework that decomposes contextual word representations into three additive components: an item component, a class component, and an idiosyncratic word-class interaction. After validating our measurement on an artificial language, we evaluate six pretrained transformers (OPT-BabyLM and Pythia families) across two linguistic categories: part of speech and grammatical role. We find that the idiosyncratic interaction term is consistently non-zero and grows substantially in deeper layers. Crucially, while abstract lexical and class axes remain nearly orthogonal, the interaction term systematically overlaps with both, allowing a probe to recover the class even after removing shared category directions. Together, our results demonstrate that LLMs do not commit exclusively to pure abstraction or exemplar storage; rather, they construct representations that are simultaneously both abstract and holistic.

1 Introduction

Lexical representations are deceptively complex: a single word often carries vastly different meanings depending on the context, yet these meanings remain recognizably related. The categorization literature has long recognized this tension, noting that assigning a novel meaning to a specific category (or word) is rarely straightforward. Simple rule-based approaches, such as rigid feature lists, quickly break down: a three-legged cat is undeniably a cat, even if it violates one of the criteria used to define the category. Ultimately, defining a lexical category in precise terms may be nearly impossible. Yet, as U.S. Supreme Court Justice Potter

Stewart famously observed in another context, we may not be able to define a category precisely, but we nevertheless “know it when [we] see it.”¹

This central paradox, that representations are simultaneously context-specific yet broadly generalizable, has fueled long-standing debates over the nature of lexical representations. The two leading frameworks offer distinct solutions: **abstraction theories** posit that humans construct generalized mental representations stripped of specific context, while **exemplar theories** argue that representations are built from distinct, context-rich memory traces.

In the present study, we investigate how Large Language Models (LLMs) navigate this paradox. Specifically, we examine the extent to which LLM lexical representations comprise abstract, generalizable features versus context-specific exemplars.

1.1 Evidence for Abstractions

Abstraction theories offer an intuitive account of the context invariance of language. At every level of language, there is substantial variability across contexts, from phonetic differences between speakers, to syntactic constructions that can take on a variety of lexical items, to the same word taking on different meanings in different contexts. The existence of abstractions accounts for the ability of humans to generalize across these contexts and recognize the same underlying representation despite this variability. Evidence for abstraction comes from linguistic productivity (Berko, 1958), semantic extension (Harmon and Kapatsinski, 2017), priming, word order preferences, and other phenomena.

Even something as deceptively simple as word order preferences, such as binomial ordering preferences, is shaped by abstract preferences (Morgan and Levy, 2015, 2016, 2024). For example, hu-

¹In *Jacobellis v. Ohio* (1964), Justice Stewart famously addressed the difficulty of defining obscenity, remarking that while formulating a precise definition might be a herculean task, he could still recognize obscene material when he saw it.

mans have strong preferences for binomial orderings such as *bread and butter* over *butter and bread*, or *bishops and seamstresses* over *seamstresses and bishops*. Those preferences are to some extent driven by abstract preferences such as a preference for male-coded words before female-coded words. Crucially, speakers’ preferences for binomials that they have never encountered before are driven by these abstract properties, so the preferences cannot be stored with any particular pair of words.

1.2 Evidence for Exemplars

Exemplar theories, in contrast, argue that abstract representations are not necessary. Instead, linguistic knowledge may be stored as context-rich memory traces whose properties vary with experience. Evidence for this view comes from both the gradient perception of phonetic cues (McMurray et al., 2008) and the representation of larger linguistic units, including multi-word phrases (Bybee, 1999; Houghton and Morgan, 2023; Houghton et al., 2025, 2026; Morgan and Levy, 2016, 2024).

More specifically, there’s evidence that humans are sensitive to within-category changes in voice onset time (McMurray et al., 2008), produce *don’t* in a more reduced manner in high-frequency phrases than in lower-frequency phrases (Bybee, 1999), have stronger word order preferences for high-frequency binomials (Morgan and Levy, 2024), and recognize *up* less well in high-frequency and high-predictability V+*up* phrases (Houghton, 2025; Kapatsinski, 2010). These findings suggest that linguistic representations retain information specific to the contexts in which their components occur, rather than being fully abstracted across contexts.

Most of the evidence above concerns sequences of words, but the same question arises within a single word. Whether regularly inflected forms are retrieved whole or assembled from a stem and an affix has been debated on much the same grounds, with frequency again playing a central role (Stemberger and MacWhinney, 2004, 1986). But the same logic applies even when there is no overt morphological combination. *Walk*, for example, occurs as both a noun and a verb. If the representation of *walk* as a verb was fully determined by a representation of *walk* together with a representation of the verb category, then nothing in that representation would be specific to that pairing. If instead some part of the representation is unique to *walk* as a verb, then the representation contains informa-

tion about a particular combination that cannot be reduced to its parts.

1.3 Present study

The evidence for both abstraction and item-specificity therefore raises a broader question about the structure of linguistic representations: is a representation of a specific linguistic construction a combination of abstract features, or does that representation also contain item-specific knowledge that is irreducible to these component parts? This is difficult to settle behaviorally, since a learner with no abstract representations at all can look like one that abstracts (Houghton and Kapatsinski, 2026), but it is a property of a representation, and unlike in humans, model representations can be examined directly (Pluth et al., 2026).

We hypothesize that linguistic representations can contain both abstract and item-specific information. We therefore decompose a word-in-class representation into three components: the item-component, the component that generalizes across different classes of the same word (the part of *walk* as a noun that is shared by *walk* as a verb), the class-component, the component that generalizes across all items of the same class (e.g., the part of *walk* as a noun that is shared by *run* as a noun, and *plant* as a noun) and what belongs to that word in that class alone (e.g., the part of the representation unique to *walk* as a noun). We first validate this decomposition in an artificial language with controlled representational structure, then apply it to six pretrained language models across part of speech and grammatical role.

Our main contributions are as follows:

- We introduce a method for decomposing word-in-context representations into lexical, grammatical, and word-class components, and validate it in an artificial language.
- Across six pretrained models, the word-class component is consistently present and grows with depth, with its size governed more by representational capacity than by language. The lexical and grammatical components occupy nearly separate axes, while the word-class component overlaps with both.
- We show that the word-class component contains information about the class itself beyond the shared class representation.

2 Experiment 1: Artificial Languages

In Experiment 1 we test our measurement on an artificial language whose features we control. This allows us to validate our measurement while simultaneously examining how different features of the language affect a model’s representations.

2.1 Dataset

In order to examine the separability of linguistic representations, we constructed an artificial language which comprises a probability distribution over a vocabulary of 600 tokens. Every observation is a combination of a **form** (one of 60), a **class** (one of 4), and a **context** (drawn from a pool of 400). The motivation for this setup is to give each word a naturalistic environment: the word occurs in a context that favors a specific class-membership (e.g., *walk* as a verb occurs in a different context than *walk* as a noun). Each form is observed 24 times in each class, every time in a different context, mirroring the patterns of human language. Consider *walk* used as a verb in one of those contexts. Four things determine the distribution over the next token. *Walk* itself contributes the same amount regardless of the class it is used in, and the verb class also contributes the same amount for every item. The third contribution is what is specific to *walk*-as-a-verb over and above those two: every form and every class has its own random vector, and this term multiplies *walk*’s by the verb’s, entry by entry. The context contributes the rest, which is noise as far as our question goes, though the model still has to encode it in order to accurately learn the language. The four are weighted, summed and passed through a softmax layer:

$$P(\text{next token} \mid f, c, x) = \text{softmax} \left(s \left[w_\alpha \text{item}(f) + w_\beta \text{class}(c) + w_\gamma \text{int}(f, c) + w_{\text{ctx}} \text{context}(x) \right] \right) \quad (1)$$

Each term is isotropically scaled to unit variance before it is weighted² so changing the weights

²Specifically, each observation consists of a form in a specific class and context, paired with that term’s raw contribution to the next-token distribution for that observation. We normalize this dataset by dividing each value in it by the total standard deviation across all observations. This is necessary because different matrices can contain the same information at different scales. For example, imagine a matrix with random numbers, and that same matrix with each number multiplied by two. The information is the same, but the weights will have a larger effect on the latter dataset.

changes the mix of the three effects without changing how much signal there is in total.

The weights w_α , w_β , and w_γ in Equation 1 control how much of the true distribution each of the three effects contributes, and we cross them factorially over $\{0, 0.5, 1, 2\}$: zero removes an effect from the language altogether, and the other three values make an effect half as strong as another, equal to it, or twice it. Similarly, w_{ctx} governs how much the context contributes, so we cross those languages with three context weights: $\{0.5, 2.0, 6.0\}$, ranging from context being a minor part of the true distribution to the dominant one. These terms are subsequently scaled to unit length, such that their squared lengths sum to 1.³ Additionally, We train a model on each of the resulting languages at hidden widths of 8, 16, 32, 128, with either a linear or a ReLU activation. The whole grid is then run with five random seeds giving 7,560 runs in all, of which 99.7% trained to convergence.

Entanglement is guaranteed under a specific subset of the languages. The structure needs 16 dimensions, so its ratio to the hidden width, which we call **crowding**, runs from 0.125 to 2.0. Below one there is room to give every component its own axis, so any entanglement was chosen; above one there are more components than dimensions to put them in, so they cannot all have their own axis and entanglement is forced (Elhage et al., 2022).

2.2 Method

All code and analyses in this paper can be found here: <https://github.com/znhoughton/representation-separability>.

2.2.1 The learner

The learner model is a multi-layer perceptron (MLP) with a single hidden layer of width d and either a linear or a ReLU activation function. The model takes as its input three pieces of information: form, class, and context. Each has its own embedding, and the three are concatenated and passed through the model which then outputs a probability distribution over the vocabulary (Figure 1). The training objective is thus to minimize cross-entropy between the model’s predicted distribution over the vocabulary and the ground-truth distribution.

³This makes the resulting languages more comparable by enforcing the same total signal (i.e., the variance of the weighted sum in Equation 1 is held constant), despite individual differences in the amount that each individual term contributes.

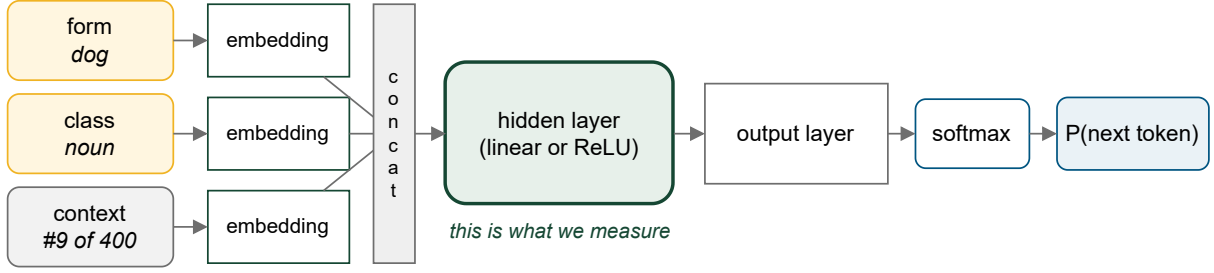


Figure 1: Model structure. Every form is observed 24 times at each class, each time in a different context; that variation across contexts is what the measurement later splits in two.

2.2.2 Decomposing a representation

We measure separability by decomposing a word-in-context representation into a lexical component, a class component, and an idiosyncratic component. The representation is the MLP’s hidden layer, which for a given form and class varies across the contexts it is observed in. Let $M[i, c] \in \mathbb{R}^d$ be the mean of those hidden states for item i used at class c . We decompose those item-by-class means by centering, so that:

$$M[i, c] = \mu + \alpha(i) + \beta(c) + \gamma(i, c), \quad (2)$$

where μ is the grand mean; $\alpha(i)$, item i ’s mean across classes minus μ , is the representation unique to the item; $\beta(c)$, the mean across items at class c minus μ , is the representation unique to the class (e.g., noun vs verb); and $\gamma(i, c)$ is the item-specific representation in a specific class, what item i at class c holds that neither marginal accounts for (Figure 2). Because every item is observed in every class with equal weight, averaging over class isolates the item effect, and averaging over items isolates the class effect, without distortion. The four terms add back to the typical representation for a given item in a given class ($M[i, c]$) exactly, and no two of them vary together across those representations, so the squared lengths of each term (excluding μ) sum to the total and we report each as a share.

Each of these averages is taken over a sample of observations that differ only in context, so each carries some of the context term of Equation 1 as noise. We remove it by dividing the observations into two even-sized splits, running the decomposition on each split separately, and estimating each component’s size as the dot product of its two split estimates, which is large when the splits agree about the component and near zero when they do not.

As a result, noise that differs between the splits is independent and cancels.

For each representation we report two measurements. The first measure is the fraction of the representation accounted for by lexical, class, and idiosyncratic information: each component’s squared length, summed over its vectors and divided by the total, so that the three fractions sum to one. The second measure quantifies the extent to which those components occupy the same directions. Specifically, we project the item vectors onto the class direction and record the fraction of their squared length that survives, and we project the interaction onto the smallest subspace containing both the item and the class vectors (i.e., the span of the two subspaces).⁴ A representation is separable if a word’s representation is the sum of an item part and a class part, each on its own axis. Separability can fail in two ways: the parts may be non-orthogonal, so that projecting out the class also removes item information, or there may be a third component that is neither, whatever its orientation. The question is accordingly not whether class information is present, but how it is represented: on a single direction shared across words, or on a separate direction for each of them.

To quantify whether a size is meaningful, we compare each against zero, which is the null because the dot product of two independent split estimates has an expected value of zero if there is no structure in the underlying representations. We redraw the split two hundred times and call a size meaningful when its confidence interval excludes zero; for components whose value is exactly zero this flags an effect about 5% of the time (Appendix A), the false-positive rate expected of a 95% confidence interval.

⁴Since there are only two classes, β spans at most one direction and its projection into the item subspace, which has many more dimensions, has high overlap by chance; we therefore do not report it.

To quantify whether an overlap is meaningful, we project the effect, held exactly as measured, onto two hundred random subspaces of the same rank, which gives the distribution of overlaps an unrelated subspace would yield. An overlap below that baseline indicates meaningful orthogonality and one above it meaningful non-orthogonality. We report each overlap as a multiple of the baseline’s median, with how many measurements fall outside its 95% interval.

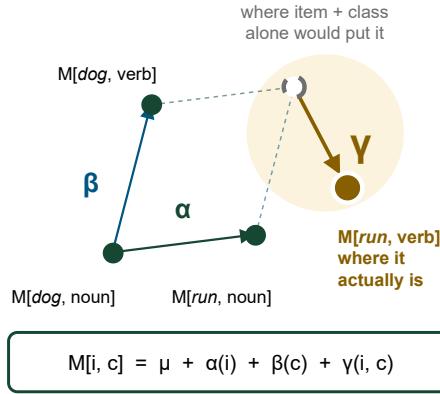


Figure 2: The four item-by-class means formed by two items in two classes. Under an item part plus a class part, three corners fix the fourth and the parallelogram closes. The arrow is $\gamma(i, c)$, the displacement between the fourth mean and the point that additive account places it at, with length $\|\gamma(i, c)\|$.

It’s important to note that the weights in Equation 1 describe the distribution the model is trained to match, while the measure decomposes the representation that the model learns, so the two sets of numbers are not on a common scale: a learner handed a language that is 85% interaction can represent it with the interaction taking a far smaller share. Appendix A reports a separate check in which components were planted directly into a representation and recovered almost exactly, which is what licenses reading the results below as facts about the learner rather than about the measure. We include runs only if at least one of the three components has a re-split interval excluding zero; where none does, the three shares divide noise and whichever lands slightly positive takes all of it.

2.3 Results

Results are shown in Figure 3; the effects of crowding are reported in Appendix B.

The interaction effect in the linear condition was zero only where the language was additive. On the other hand, across the 2,880 runs whose language

contained an interaction, the linear model’s interaction interval excludes zero in 36% of them, against 77% for ReLU. In contrast, where no interaction was planted the rates were 2% and 13%. The linear model has no parameter depending on the form and the class together beyond a linear combination of them, yet represents an interaction anyway. One possible explanation for this is that the model can still learn context-specific representations for different items, by learning that a specific item in a specific class occurs with a specific context (and then subsequently representing that information within the context weights). In the models with a ReLU activation function, surprisingly we see an interaction effect represented in the model even in cases where the language is additive. One potential explanation is that the model only ever encounters words in specific contexts; the additive structure of the language is visible across the corpus rather than in any single observation, and the model does not seem to abstract away the interaction completely.

For models with larger dimensionality, the added room seems to be allocated toward representing the interaction. From a hidden width of 16 to a width of 128, the interaction’s share of the item-by-class structure rises from 0.002 to 0.068, while the item effect falls from 0.58 to 0.47 and the class effect holds steady at around 0.43. In the linear model the same width is spent elsewhere: the item effect rises from 0.61 to 0.72 while the interaction stays at 0.006.

3 Experiment 2: Language Models

Experiment 1 established that if the language a learner model encounters is not purely additive, the model will represent the language in a non-separable manner, with the extent governed by the size and crowding of the model. Experiment 2 applies the same decomposition to the contextual representations of pretrained language models, across two linguistic categories: part of speech and grammatical role.

3.1 Dataset

In order to examine the separability of linguistic representations in models, we examined two linguistic categories: part of speech and grammatical role. For both, we concatenate six general-domain English Universal Dependencies treebanks (Nivre et al., 2020): EWT (Silveira et al., 2014), GUM (Zeldes, 2017), LinES (Ahrenberg, 2007),

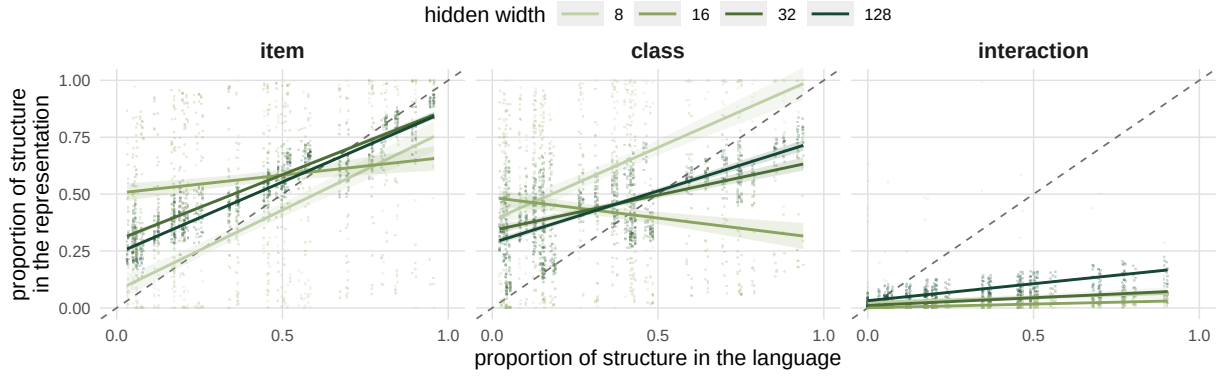


Figure 3: What the model represents of each component, against how much of that component the ground-truth language held, under ReLU. Each faint point is one run. Lines are least-squares fits through the runs at each hidden width, shaded with a 95% confidence band on the fit. The dashed diagonal is where a representation would sit had it reproduced the language’s composition exactly. Runs in which none of the three components had a re-split 95% interval excluding zero are not plotted, since the three shares then divide noise.

ParTUT (Sanguinetti and Bosco, 2015), GENTLE (Aoyama et al., 2023), and PUD (Zeman et al., 2017), giving 41,095 sentences and 706,323 tokens. We measure two model families at three matched scales: OPT-BabyLM, trained on developmentally plausible quantities of child-available language (Houghton et al., 2026; Warstadt et al., 2023), and Pythia, trained on the Pile (Biderman et al., 2023), at 125M/160M, 350M/410M, and 1.3B/1.4B parameters, with widths of 768, 1024, and 2048. Appendix C gives parameter counts and training-token estimates.

3.2 Stimuli

Our stimuli comprise two linguistic categories: part of speech (NOUN versus VERB) and grammatical role (*nsubj* versus *obj*, restricted to nouns). For each category we select items that occur in both of its classes, such as *walk*, which occurs as both a noun and a verb. An item is a single inflected word form, not a lemma, so different inflections of one lemma are separate items: *walk* and *walking* are both items, each included only if it is itself attested in both classes. Pooling them under one lemma would collect *walk*, *walks*, *walked*, and *walking* into a single item, and the two classes would then be read at different tokens, which the embedding layer could then separate before any contextualization has occurred. Capitalization is ignored when occurrences are grouped, but not in the text the models read, so a sentence-initial *Walk* counts toward the item *walk* and is tokenized as it appears there.⁵

⁵OPT-BabyLM’s tokenizer is uncased, so this choice has no effect for that family of models. On the other hand, Pythia’s

An item is included only if it occurs at least ten times in each class; forms occurring in only one are excluded, which makes part of speech the more restrictive of the two. This yields 78 items for part of speech and 49 for grammatical role, with median counts per class running from 10 to 36 (Table 10), inside the range we validated for the measurement (Appendix A).

3.3 Method

The decomposition, the two reported quantities, and the replication logic are identical to Experiment 1. For every token we take the hidden state of the word’s **last subword** at every layer of each model (using the model’s own tokenizer).⁶

Transformer hidden states contain a few *rogue* dimensions whose variance exceeds the rest by two to three orders of magnitude (Timkey and van Schijndel, 2021), and these would otherwise dominate any subspace estimated from the raw scatter, so we divide each dimension by its standard deviation before decomposing.

3.4 Results

The results for the final layer are reported in Table 11; the layer-by-layer results are shown in Figure 4.

The item effect is the largest component in both

tokenizer is not uncased; however, lowercasing the input would be the larger distortion, since a sentence-initial word is capitalized in written English which the model was trained on.

⁶Almost every item is a single subword, so this choice has little bearing on the results: zero items are split into multiple subword tokens under Pythia’s tokenizer, and only eight are split into multiple subword tokens under OPT-BabyLM’s tokenizer.

categories, running from 0.656 to 0.829 for part of speech and 0.828 to 0.898 for grammatical role. What differs between them is how the remainder divides. For grammatical role the class effect, at 0.047 to 0.078, is comparable to the interaction at 0.044 to 0.094; for part of speech the interaction, at 0.105 to 0.245, runs well above the class effect at 0.057 to 0.099.

Both the class effect and the interaction grow stronger through the layers of the model. The interaction is essentially absent at layer 0 in both categories, between 0.0009 to 0.0068 across every category and model, and rises to between 0.0438 to 0.2450 (Figure 4, top row). This is not surprising since at the first layer, the model has yet to integrate enough context into the representation to commit to the class of the representation. What the network builds over those layers is also not an artifact of where a word sits in the sentence: zeroing OPT-BabyLM’s learned absolute position embeddings leaves the class structure more word-specific rather than less, and Pythia, which has no such embeddings to remove, shows the interaction at least as strongly (Appendix G).

Turning to how those components are oriented, the class direction carries almost none of the item effect. At the deepest layer, item-class overlap runs 0.0032 to 0.0067 for part of speech and 0.0025 to 0.0062 for grammatical role, so projecting out the class direction leaves more than 99% of the item effect intact. While both overlaps are technically less orthogonal than a random subspace, the overlap itself is tiny, so the two are near-orthogonal rather than exactly orthogonal.

In contrast to the main effects, the interaction is not orthogonal to class or item. Across the six models, 0.096 to 0.239 of the interaction for part of speech is in the span of the two main effects, and for grammatical role 0.084 to 0.179 is, against nulls whose upper ends run 0.023 to 0.058 and 0.014 to 0.038 respectively. The smallest of these overlaps exceeds the largest of the nulls, so the pattern holds in every model. In other words, the small departure from orthogonality between item and class becomes a large one for the interaction.

4 Experiment 3: Probing the Interaction

Experiment 2 shows that the interaction is large and that it replicates across independent splits of a word’s occurrences, but neither shows what information is contained within the interaction. An

item-by-class term captures any reliable difference between a word’s uses in the two classes, and some of that difference may be correlated with the context, since subjects appear in different constructions from objects.

4.1 Method

We test whether the interaction encodes class information with a probe. If γ encodes the class, the class should be recoverable from the part of a token’s representation that is neither the item effect nor the shared class direction, and we remove both explicitly. All probing happens *within* a single word, so we ask whether *walk*’s noun uses can be told from *walk*’s verb uses; the item effect is constant within a word and so cannot contribute. We then estimate β as the difference between the two class means pooled over all items, normalize it, and remove that component from every token, so that whatever separates the classes afterwards is not the direction they share.

What remains is the interaction: the part of a word of a specific class that is neither a fact about the word nor a fact about the class. We fit a cross-validated logistic probe to it, five folds, with classes balanced by subsampling within each word so that chance is 0.50, and average the resulting accuracies over words, with a word contributing only if it has at least eight tokens in each class. Some words rest on few tokens, so the mean carries a percentile bootstrap interval over words. The whole procedure is then repeated with the class labels permuted within each word, which preserves every word’s token count and class balance while destroying the labeling, and that null is where a probe reading its own cross-validation folds would land (Hewitt and Liang, 2019). We run this at the deepest layer of the largest model in each family, for both categories.

4.2 Results

Results are reported in Table 1.

For both families of LLMs across both linguistic categories, the class is recoverable from the interaction at well above chance while the permuted null sits on chance, reaching 0.94 for part of speech and 0.84 for grammatical role, with bootstrap intervals excluding the nulls in all four cases. The null landing on chance is what makes those accuracies interpretable, since it is the same probe, the same folds, and the same word-level averaging with only the labeling destroyed.

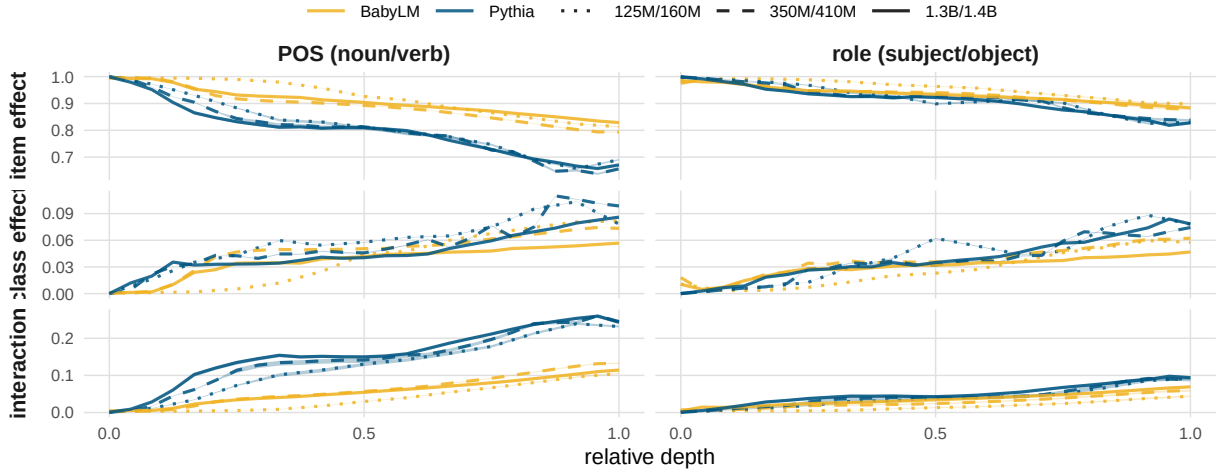


Figure 4: The three components through the network, for every model and linguistic category. Relative depth runs from the static embedding layer (0) to the final block (1). Rows are the three components’ shares of the item-by-class structure, which sum to one within each model and layer. Each line is one model: color distinguishes the training corpus, linetype the three matched widths. The shaded band is that model’s re-split 95% interval.

class	model	accuracy	null
part of speech	BabyLM 1.3B	0.93 [0.91, 0.94]	0.52 [0.49, 0.55]
part of speech	Pythia 1.4B	0.94 [0.93, 0.95]	0.50 [0.48, 0.53]
grammatical role	BabyLM 1.3B	0.85 [0.82, 0.87]	0.50 [0.47, 0.53]
grammatical role	Pythia 1.4B	0.84 [0.82, 0.87]	0.52 [0.49, 0.55]

Table 1: Recovering the class from the interaction with a probe, at the deepest layer of the largest model in each family. The item is held fixed and the shared class direction is projected out, so neither main effect can contribute. *Accuracy* is the mean over them; *null* repeats the probe with the class labels permuted within each form. Brackets are percentile bootstrap intervals over forms, and chance is 0.50.

The probe accuracies follow the same ordering across categories as the interaction sizes (Table 11), providing converging evidence from the geometric and functional measures.

The results demonstrate that γ carries class information rather than generic correlated context, since generic context would not survive holding the word fixed and removing the shared axis. It does not show that γ is *only* the label: context correlated with the class within a word survives both removals, so the interaction remains an upper bound.

5 Discussion and Conclusion

The present study examined whether a word’s representation in a language model is exemplar-like or abstract and found that it contains elements of both. We decomposed a word’s representation in a class into an item effect, a class effect, and a

word-class interaction. The first two capture information shared across classes or words, respectively, while the interaction captures information specific to a particular word-class combination. Across six pretrained language models and two linguistic categories, this interaction was present in every model and category, nearly absent at the embedding layer, and grew throughout the network. The abstract components were nearly orthogonal, whereas the interaction overlapped with both, indicating that abstract and item-specific information are represented together rather than on separate axes.

Overall, the central result is not simply that language-model representations are sensitive to context, but that this context sensitivity is organized around particular word-class combinations. Language-model representations therefore combine abstraction with item-specific structure: they generalize across words and classes while retaining information about particular combinations.

Limitations

Our evaluation is restricted to English and relies on two linguistic categories tested across two model families (OPT-BabyLM and Pythia) at three parameter scales. While these choices cover a range of architectures and linguistic phenomena, broader typological, categorical, and architectural validation remains a direction for future work.

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A Validating the Measure

Experiment 1 plants structure in a language and measures a trained model’s representation. However, that approach cannot distinguish a failure of the measure from the model representing the language differently than the planted structure. Here we remove that ambiguity by planting μ , α , β , and γ directly in a representation, with no model or training, and passing the result through the same analysis used for the language models. Ground truth therefore exists at the level where the measure operates.

Specifically, we generated representations with known values of μ , α , β , and γ , varying the number of items, classes, representation dimensions, observations per pairing, and noise level across the range used in the experiments. Components were centered to satisfy their defining constraints and rescaled to predetermined shares of the total squared length. The grid included the same factorial Experiment 1 runs, mapped into share space, the three compositions measured in Experiment 2, item counts from 27 to 180, two and four classes, widths from 8 to 2048, 10, 24, and 100 observations per item and class, three noise levels, and five seeds. We also generated orthogonal and known non-orthogonal item and class effects: the orthogonal condition draws α and β independently, while the non-orthogonal condition places a known fraction of β along α ’s leading singular direction. In total, this gave 252,000 runs.

At 24 observations per item and class, the measured interaction matches the planted one across the whole range (Table 2), and is flagged in essentially every run where anything was planted and about one in twenty where nothing was.

planted	runs	measured	flagged
0.000	6300	0.0000 [0.0000, 0.0094]	5%
0.030	900	0.0323 [0.0199, 0.0418]	100%
0.048	1800	0.0495 [0.0385, 0.0688]	100%
0.056	1800	0.0573 [0.0458, 0.0918]	100%
0.059	1800	0.0607 [0.0495, 0.1041]	100%
0.100	1800	0.1007 [0.0911, 0.1089]	100%
0.111	900	0.1154 [0.1031, 0.1279]	100%
0.167	1800	0.1708 [0.1594, 0.2204]	100%
0.190	1800	0.1937 [0.1825, 0.2736]	100%
0.200	1800	0.2030 [0.1926, 0.2981]	100%
0.240	900	0.2416 [0.2327, 0.2488]	100%
0.333	900	0.3396 [0.3263, 0.3610]	100%

Table 2: Recovery at 24 observations per item and class in 768 or more dimensions, pooled over noise levels, item counts, and widths. Brackets are the 2.5th and 97.5th percentiles across runs.

All three effects read essentially zero when nothing is planted, at every sample size, and each is flagged at approximately the nominal 5% rate (Table 3).

effect	observations	median	95th pct	flagged
class effect	10	0.00000	0.0081	6.1%
class effect	24	0.00000	0.0034	6.5%
class effect	100	0.00000	0.0009	6.3%
interaction	10	0.00009	0.0644	6.2%
interaction	24	0.00004	0.0309	5.9%
interaction	100	0.00002	0.0083	4.9%
item effect	10	0.00007	0.0568	4.1%
item effect	24	0.00003	0.0255	3.6%
item effect	100	0.00000	0.0064	5.0%

Table 3: What the measure reports when an effect is planted at exactly zero, by observations per item and class, pooled over every other axis of the validation grid.

The false-positive rate varies somewhat with representation width and component, particularly at small widths, but remains below 10% throughout the validation grid (Table 4). The language-model analyses

use $d \geq 768$, where the rates are closer to the nominal level.

width d	item	class	interaction
8	3%	14%	7%
16	7%	3%	5%
32	4%	4%	10%
128	6%	4%	5%
768	2%	8%	3%
1024	3%	5%	3%
2048	0%	8%	8%

Table 4: How often a component planted at exactly zero is flagged, by width and by effect.

width d	measured	analytic chance
8	0.17501 [0.06900, 0.40018]	0.25000
16	0.10678 [0.03744, 0.20884]	0.12500
32	0.05609 [0.01864, 0.10454]	0.06250
128	0.01714 [0.00450, 0.02611]	0.02344
768	0.00261 [0.00079, 0.00472]	0.00391
1024	0.00217 [0.00065, 0.00330]	0.00293
2048	0.00096 [0.00031, 0.00165]	0.00146

Table 5: Measured item-class overlap when the two effects are planted orthogonal, at 24 observations per item and class. *Measured* is the median over the runs at that width with the 2.5th and 97.5th percentiles in brackets. The *chance* column is the analytic level of Appendix E.

With orthogonal item and class effects, the measured overlap tracks the chance level across representation widths (Table 5). When a known fraction of the class direction is placed inside the item subspace, the measured overlap rises above chance and is detected in 89% of runs at $d = 2048$ (Table 6). Thus, the measure recovers both the expected chance overlap under orthogonality and genuine non-orthogonality.

width d	measured	chance	detected
8	0.1688 [0.0824, 0.3998]	0.12500	9%
16	0.0951 [0.0414, 0.2026]	0.06250	22%
32	0.0593 [0.0225, 0.1119]	0.03125	36%
128	0.0213 [0.0062, 0.0517]	0.00781	63%
768	0.0065 [0.0011, 0.0298]	0.00260	83%
1024	0.0056 [0.0008, 0.0331]	0.00098	82%
2048	0.0036 [0.0004, 0.0290]	0.00049	89%

Table 6: Measured item-class overlap when a fraction of the class direction is planted INSIDE the item subspace (non-orthogonal), at 24 observations per item and class. *Measured* is the median over the runs at that width with the 2.5th and 97.5th percentiles in brackets; *chance* is the analytic random-orientation level; *detected* is the fraction of runs whose overlap falls outside that null’s 95% interval.

B Crowding and Orthogonality

Whether the item and class effects are orthogonal depends in part on the capacity of the representation. We therefore hold the language fixed while varying hidden width, and measure overlap as a function of crowding: the rank required by the structure divided by the available width (Figure 5; Table 7).

Under ReLU, overlap remains below chance when the representation has sufficient capacity, but rises above chance once the required rank approaches or exceeds the available width. The linear condition shows a similar increase in raw overlap but remains near its chance level throughout. Thus, compression alone does not imply entanglement: a representation can encode the same structure in fewer directions without causing the item and class effects to become aligned.

At low widths, the representations also vary substantially across runs, with this variability decreasing as width increases. For example, two runs of the same language differ by 0.47 in the item effect and 0.52 in the class effect at width 8, compared with 0.08 and 0.07 at width 32. This variability reflects the different ways models can distribute information when representational capacity is limited.

Two properties of the language also affect the composition of the item-by-class structure (Table 8). Increasing the contribution of context reduces the interaction share, while increasing the planted item share produces a corresponding increase in the measured item share.

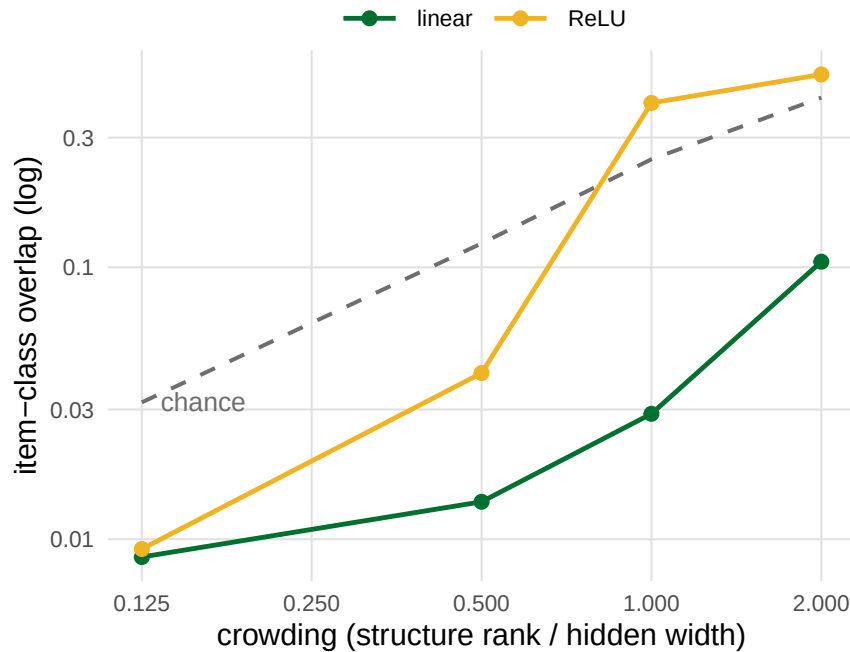


Figure 5: Item-class overlap against crowding, by activation, over the 7,560 artificial-language runs. Crowding is the rank the structure needs divided by the hidden width. The dashed line is the median upper end of the null for an arbitrarily oriented effect. Both axes are logarithmic.

condition	crowding	runs	interaction	item-class overlap	interaction overlap
linear	0.125	405	0.008 [0.002, 0.017] (241/405)	0.0086 (0.4×, 333/405)	—
linear	0.500	405	0.004 [0.001, 0.019] (189/405)	0.0137 (0.2×, 398/405)	—
linear	1.000	405	0.002 [0.000, 0.024] (57/405)	0.0289 (0.2×, 348/393)	—
linear	2.000	396	0.002 [0.000, 0.556] (111/396)	0.1049 (0.4×, 188/319)	—
ReLU	0.125	405	0.083 [0.036, 0.186] (405/405)	0.0092 (0.6×, 225/405)	0.087 (1.3×, 316/405)
ReLU	0.500	405	0.037 [0.003, 0.073] (367/405)	0.0408 (0.7×, 231/405)	0.142 (0.7×, 214/332)
ReLU	1.000	405	0.008 [0.000, 0.046] (237/405)	0.4022 (6.9×, 271/394)	0.230 (1.1×, 40/97)
ReLU	2.000	315	0.003 [0.000, 0.429] (189/315)	0.5119 (5.1×, 122/226)	0.582 (2.4×, 40/77)

Table 7: Interaction size and the two overlaps by activation and crowding, over the 7,560 artificial-language runs. Crowding is the rank the structure needs divided by the hidden width; each crowding value corresponds to one width, from 128 down to 8. *Interaction* is the median share of the representation over the runs in the row, their interquartile range in brackets, and how many have a re-split 95% interval excluding zero. The overlap columns give the median fraction surviving the projection, then in parentheses how many times its own null median that is and how many runs fall outside that null. The linear arm’s interaction stays below the 2% floor at which orientations are reported, so its last column is empty.

property	level	runs	item	class	interaction
context in the language	under 30%	720	0.451	0.482	0.027
	30 to 70%	572	0.494	0.460	0.016
	over 70%	745	0.505	0.459	0.010
item against class	item < class	694	0.330	0.605	0.020
	item = class	674	0.492	0.447	0.019
	item > class	669	0.643	0.311	0.014

Table 8: How two other properties of the artificial language move the composition of the item-by-class structure, under ReLU, pooled over widths. Rows within a property partition the same runs. Values are medians of the three shares over the runs in the row.

family	model	params	width	layers	training data	tokens
BabyLM	opt-babylm-125m	125M	768	12	BabyLM (child-available)	~3B
	opt-babylm-350m	350M	1024	24	BabyLM (child-available)	~3B
	opt-babylm-1.3B	1.3B	2048	24	BabyLM (child-available)	~3B
Pythia	pythia-160m	160M	768	12	the Pile	~300B
	pythia-410m	410M	1024	24	the Pile	~300B
	pythia-1.4b	1.4B	2048	24	the Pile	~300B

Table 9: The six models, paired by width. Training-token counts are approximate; the OPT-BabyLM models train for 20 epochs over a 100M-word corpus, which is roughly 150M tokens, so their effective token exposure is about twenty times that. *Layers* counts the transformer blocks, and the measure is applied to each block’s output plus the embedding layer.

The six models are listed in Table 9. The two families are matched on width rather than on parameter count, which is what the measure is sensitive to: the chance level for overlap is the class subspace rank over the width (Appendix E), so a comparison across models of different width has to be read against different chance levels. Pairing at 768, 1024, and 2048 removes that from the family comparison.

They differ in two ways that matter elsewhere in the paper. The training corpora differ by more than two orders of magnitude in size, which is the manipulation of interest. And they position tokens differently: Pythia uses rotary embeddings, applied inside attention, while OPT learns absolute position embeddings added at the input. Appendix G uses that difference to rule out absolute position as the source of the interaction.

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D Stimulus Counts

class	item types	median tokens
part of speech (noun / verb)	78	18 / 36
grammatical role (subject / object)	49	10 / 15

Table 10: Stimuli for the two linguistic categories, with the two classes of each in parentheses. *Item types* is how many word forms occur in both of them at least ten times; forms occurring in only one are excluded. *Median tokens* is how often a typical form occurs in each of its two classes, in the order they are listed.

E The Decomposition

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E.1 The item-by-class means

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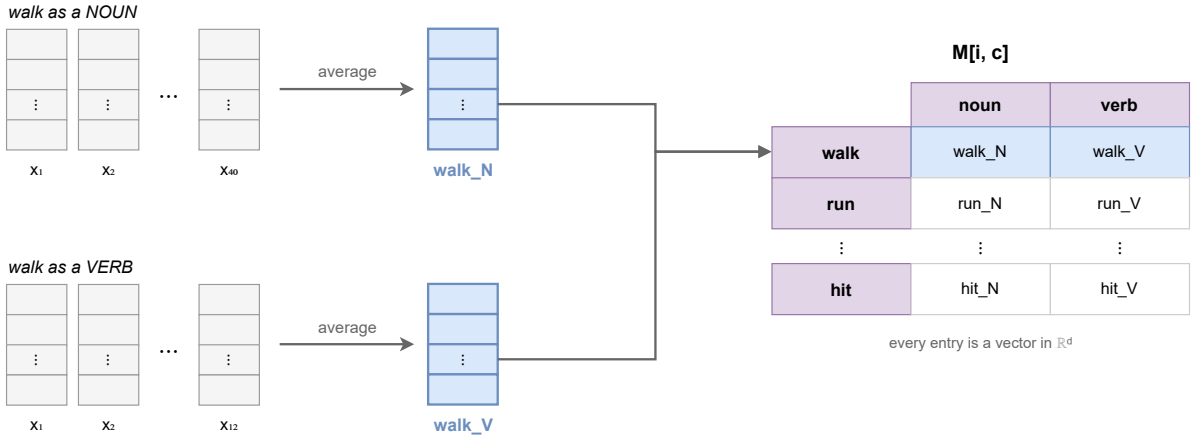


Figure 6: Each word in each class is observed in many contexts, and those observations are averaged within the pairing to give one of the item-by-class means. Every one of them is a vector in $\mathbb{R}^d$, and every item appears in every class.

A word used in a given class is observed in many contexts, and the representation differs a little in each. Averaging those observations within the pairing gives one of the item-by-class means: for items $i \in \{1, \dots, I\}$ crossed with classes $c \in \{1, \dots, C\}$, $M[i, c] \in \mathbb{R}^d$ is the mean representation of item i in class c (Figure 6). An item observed in only one class is dropped, as is an item with fewer than 10 tokens in either class, so the set is complete and every mean rests on a comparable number of observations. Because every item appears at every class, the means arrange into I by C positions, and the rest of this appendix calls that arrangement the grid when what matters is the positions rather than the vectors sitting at them. The variation across contexts that the averaging sets aside is not discarded; it is what the two splits of Section 2.2.2 are drawn from.

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E.2 Sizes

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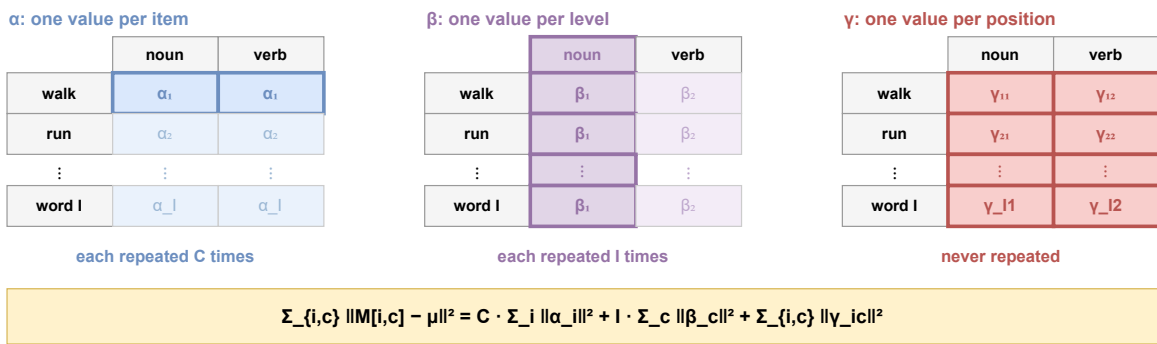


Figure 7: How many positions of the grid each component occupies. α_i does not depend on the class, so it occupies all C positions of its row; β_c does not depend on the item, so it occupies all I positions of its column; γ_{ic} takes its own value at every position and is never repeated.

Since the effects are centered along their respective axes, their cross-terms cancel when summed over the full grid. Thus the total squared length partitions exactly:

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$$\sum_{i,c} \|M[i, c] - \mu\|^2 = C \sum_i \|\alpha_i\|^2 + I \sum_c \|\beta_c\|^2 + \sum_{i,c} \|\gamma_{ic}\|^2,$$

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Sizes are reported as shares of the total. All three terms scale linearly in I , so the shares do not depend on how many items there are and are comparable across categories of different size.

The above decomposition, however, ignores the role of context. In order to estimate how the three components and context vary in size, we can instead decompose each representation into its pairing mean and its deviation from that mean, giving the standard within–between partition:

$$\sum_x \|x - \bar{x}\|^2 = \underbrace{\sum_{i,c} \sum_{x \in (i,c)} \|x - \hat{M}[i, c]\|^2}_{\text{within a pairing}} + \underbrace{\sum_{i,c} n_{ic} \|\hat{M}[i, c] - \bar{x}\|^2}_{\text{between pairings}}.$$

The between-pairing term is the variation partitioned into item, class, and item-by-class components above, while the within-pairing term captures variation across contexts. The resulting context variance and the α , β , and γ variance components together partition the variance of the original representations.

E.3 The overlaps and their chance level

The component sizes describe how much variation each component accounts for, but not whether the components occupy the same directions in $\mathbb{R}^d$. To measure their overlap, we project each component into a target subspace and ask how much of its squared length is retained. Specifically, we project α into the subspace spanned by β , β into the subspace spanned by α , and γ into the subspace jointly spanned by α and β . If S is an orthonormal basis for the target subspace and V is the component being projected, the overlap is

$$\frac{\|VS\|_F^2}{\|V\|_F^2}$$

Thus, an overlap of zero means that the component is orthogonal to the target subspace, whereas an overlap of one means that it lies entirely within it. Because the projection uses squared lengths, opposite directions contribute equally rather than canceling. We use the leading singular vectors of each target component to define its subspace, with the number of directions determined by its participation ratio. The number of directions in the target subspace therefore determines its chance overlap under random orientation.

The chance level for overlap is r/d , where r is the dimensionality of the target subspace and d is the full representation dimension. A randomly oriented component therefore has an expected fraction r/d of its squared length in the target subspace. At $d = 2048$, for example, the chance level is 0.0005 for a one-dimensional class subspace. We therefore interpret observed overlaps relative to this chance level.

F Deepest-Layer Measurements

Table 11 gives the deepest-layer values behind the curves of Figure 4, together with the null each overlap is read against.

class	model	item-class overlap	interaction overlap	class effect	interaction
part of speech	BabyLM 125M	0.0056 (4.4×)	0.172 (3.5×)	0.082 [0.081, 0.082]	0.105 [0.104, 0.106]
	BabyLM 350M	0.0061 (6.4×)	0.157 (4.0×)	0.073 [0.073, 0.073]	0.133 [0.131, 0.134]
	BabyLM 1.3B	0.0032 (6.7×)	0.096 (4.5×)	0.057 [0.057, 0.057]	0.114 [0.113, 0.116]
	Pythia 160M	0.0058 (4.6×)	0.230 (4.2×)	0.078 [0.078, 0.078]	0.232 [0.229, 0.234]
	Pythia 410M	0.0067 (7.1×)	0.239 (6.1×)	0.099 [0.098, 0.099]	0.245 [0.242, 0.248]
	Pythia 1.4B	0.0056 (11.6×)	0.209 (9.8×)	0.086 [0.086, 0.086]	0.243 [0.240, 0.246]
grammatical role	BabyLM 125M	0.0038 (3.1×)	0.121 (3.6×)	0.058 [0.058, 0.058]	0.044 [0.042, 0.046]
	BabyLM 350M	0.0041 (4.5×)	0.112 (4.4×)	0.062 [0.062, 0.063]	0.059 [0.056, 0.061]
	BabyLM 1.3B	0.0025 (5.4×)	0.084 (5.7×)	0.047 [0.047, 0.047]	0.069 [0.066, 0.072]
	Pythia 160M	0.0049 (3.9×)	0.179 (5.3×)	0.078 [0.078, 0.078]	0.092 [0.088, 0.096]
	Pythia 410M	0.0047 (5.0×)	0.160 (6.6×)	0.074 [0.074, 0.075]	0.087 [0.084, 0.091]
	Pythia 1.4B	0.0062 (12.5×)	0.141 (11.1×)	0.078 [0.078, 0.079]	0.094 [0.090, 0.098]

Table 11: Deepest-layer measurements, one row per model. The two overlaps are the fraction of the item effect lying in the class subspace and the fraction of the interaction lying in the span of the two main effects, each followed in parentheses by how many times its own rotation null’s median it is. That null holds the effect fixed and rotates the target subspace at random 200 times. The class effect and the interaction are shares of the item-by-class structure, with each model’s re-split 95% interval in brackets.

G Positional Embeddings Are Not the Source

One architectural explanation of the interaction has to be ruled out before it can be read as a fact about lexical representations. The BabyLMs add a *learned absolute position embedding* at the input. The two classes of a linguistic category are not positionally interchangeable in a sentence, since subjects precede objects and nouns and verbs occupy different slots, so a representation that varies with position varies with the class. If that is what the decomposition is picking up, the interaction would be an artifact of where words sit rather than of how they are represented.

We re-extracted every OPT-BabyLM model with the positional embeddings zeroed out and measured it identically. The ablation was verified per model rather than assumed: with the embedding zeroed, the same token at two different offsets must give identical layer-0 states, and it does, in every model (drift 0.0 to machine precision). Pythia has no such module, so there is nothing to zero there.

We measure the class effect’s share of class-relevant structure which is the class effect divided by the sum of the class effect and the interaction, which we call the *portable fraction*. If positions were manufacturing the interaction, removing them should leave a larger share of what distinguishes the two classes in a direction shared across words. Absolute sizes cannot be compared here: zeroing the embeddings in a model that was trained with them is a heavy intervention, and it shrinks every component at once, the class effect included.

model	positions	class	interaction	portable fraction
opt-babylm-125m	zeroed	0.011	0.033	0.254
opt-babylm-125m	intact	0.081	0.105	0.436
opt-babylm-350m	zeroed	0.012	0.045	0.206
opt-babylm-350m	intact	0.073	0.134	0.353
opt-babylm-1.3B	zeroed	0.008	0.034	0.188
opt-babylm-1.3B	intact	0.057	0.116	0.329

Table 12: Trained OPT-BabyLM at the deepest layer, with absolute position embeddings intact and zeroed. Sizes are shares of the item-by-class structure and the portable fraction is the class effect divided by the sum of the class effect and the interaction.

Removing the position embeddings makes these representations *less* separable, not more. The portable fraction falls in every model, from 0.439 to 0.264, from 0.468 to 0.253, and from 0.333 to 0.198 (Table 12). Whatever the absolute position embedding contributes, it is contributing to the part of class structure that *is* carried by a shared direction. Take it away and what remains is more word-specific, not less. The interaction is not an artifact of position.